

Foundations of Recommender system

[Recommender system]

1.0 Content Block

One approach to the design of recommender systems that has wide use is collaborative filtering. Collaborative filtering is based on the assumption that people who agreed in the past will agree in the future, and that they will like similar kinds of items as they liked in the past. The system generates recommendations using only information about rating profiles for different users or items. By locating peer users/items with a rating history similar to the current user or item, they generate recommendations using this neighborhood. This approach is a cornerstone for e-commerce sites that analyze the purchasing patterns of thousands of users to suggest what you might like. Collaborative filtering methods are classified as memory-based and model-based. A well-known example of memory-based approaches is the user-based algorithm, while that of model-based approaches is matrix factorization (recommender systems). A key advantage of the collaborative filtering approach is that it does not rely on machine analyzable content and therefore it is capable of accurately recommending complex items such as movies without requiring an "understanding" of the item itself. Many algorithms have been used in measuring user similarity or item similarity in recommender systems. For example, the k-nearest neighbor (k-NN) approach and the Pearson Correlation as first implemented by Allen. When building a model from a user's behavior, a distinction is often made between explicit and implicit forms of data collection. Examples of explicit data collection include the following:

2.0 Content Block

One of the events that energized research in recommender systems was the Netflix Prize. From 2006 to 2009, Netflix sponsored a competition, offering a grand prize of \$1,000,000 to the team that could take an offered dataset of over 100 million movie ratings and return recommendations that were 10% more accurate than those offered by the company's existing recommender system. This competition energized the search for new and more accurate algorithms. On 21 September 2009, the grand prize of US\$1,000,000 was given to the BellKor's Pragmatic Chaos team using tiebreaking rules. The most accurate algorithm in 2007 used an ensemble method of 107 different algorithmic approaches, blended into a single prediction. As stated by the winners, Bell et al.:

3.0 Content Block

Recommend systems are essential for modern e-commerce platforms, playing a key role in improving the customer experience and increasing sales. These systems analyze customer data to provide personalized product suggestions, helping users discover items they might not have found on their own. A study by J. Leskovec et al. highlighted that such systems are crucial when an individual needs to choose from a potentially overwhelming number of items that a service may offer. E-commerce recommenders typically use a combination of filtering techniques to generate these suggestions. Collaborative filtering is a core method, recommending products based on the purchasing and Browse habits of similar users. Another widely used approach is content-based filtering, which recommends items with similar attributes to those a user has previously shown interest in. Many e-commerce platforms use a hybrid approach, combining these techniques to create more accurate and diverse recommendations, which helps to address issues like the "cold start" problem for new users or products. These systems are implemented in several ways across e-commerce sites to maximize their effectiveness at different stages of the shopping process:

4.0 Content Block

Predictive accuracy is substantially improved when blending multiple predictors. Our experience is that most efforts should be concentrated in deriving substantially different approaches, rather than refining a single technique. Consequently, our solution is an ensemble of many methods. Many benefits accrued to the web due to the Netflix project. Some teams have taken their technology and applied it to other markets. Some members from the team that finished second place founded Gravity R&D, a recommendation engine that's active in the RecSys community. 4-Tell, Inc. created a Netflix project-derived solution for ecommerce websites. A number of privacy issues arose around the dataset offered by Netflix for the Netflix Prize competition. Although the data sets were anonymized in order to preserve customer privacy, in 2007 two researchers from the University of Texas were able to identify individual users by matching the data sets with film ratings on the Internet Movie Database (IMDb). As a result, in December 2009, an anonymous Netflix user sued Netflix in *Doe v. Netflix*, alleging that Netflix had violated United States fair trade laws and the Video Privacy Protection Act by releasing the datasets. This, as well as concerns from the Federal Trade Commission, led to the cancellation of a second Netflix Prize competition in 2010.

5.0 Content Block

Television As the connected television landscape continues to evolve, search and recommendation are seen as having an even more pivotal role in the discovery of content. With broadband-connected devices, consumers are projected to have access to content from linear broadcast sources as well as internet television. Therefore, there is a risk that the market could become fragmented, leaving it to the viewer to visit various locations and find what they want to watch in a way that is time-consuming and complicated for them. By using a search and recommendation engine, viewers are provided with a central 'portal' from which to discover content from several sources in just one location.

6.0 Content Block

Content-based filtering Another common approach when designing recommender systems is content-based filtering. Content-based filtering methods are based on a description of the item and a profile of the user's preferences. These methods are best suited to situations where there is known data on an item (name, location, description, etc.), but not on the user. Content-based recommenders treat recommendation as a user-specific classification problem and learn a classifier for the user's likes and dislikes based on an item's features. In this system, keywords are used to describe the items, and a user profile is built to indicate the type of item this user likes. In other words, these algorithms try to recommend items similar to those that a user liked in the past or is examining in the present. It does not rely on a user sign-in mechanism to generate this often temporary profile. In particular, various candidate items are compared with items previously rated by the user, and the best-matching items are recommended. This approach has its roots in information retrieval and information filtering research. To create a user profile, the system mostly focuses on two types of information: